Towards Next Generation Risk Assessment of PFAS. Development and evaluation of a mechanistic human PBK model for PFOA.
Step 1

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Abstract

Per- and polyfluoroalkyl substances (PFAS) represent a substantial chemical class, comprising thousands of structurally diverse chemicals, which only have in common one fully fluorinated methyl or methylene carbon atom. PFAS are ubiquitous environmental pollutants of anthropogenic nature and therefore of potential risk to human health. Human toxicological risk assessment of PFAS is challenging, due to their sheer number, structural diversity and paucity of the relevant data required to identify and characterize them. A solution to this challenge lies in the application of read-across approach, which involves the extrapolation of information from a data-rich PFAS to make predictions about a similar data-poor one. One example of this approach could be the application of a Physiologically Based Kinetic (PBK) model from PFOA, to inform the development of PBK models for PFASs with analogous physicochemical properties. Despite the availability of several PBK models for PFOA, all of them rely on PFOA-specific animal-sourced data and are optimised to correctly predict the long half-lives observed in humans. This hinders their use in the read-across scope, where PBK models should be as mechanistic as possible, with key toxicokinetic processes are described based on either in vitro or in silico studies. The present study introduces a human mechanistic PBK model for PFOA, that uses only in vitro derived input parameters. The uptake and elimination of PFOA are predicted using in vitro to in vivo extrapolation principles, while its tissue

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[1] This is the first author footnote.

[2] Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

[3] Yet another author footnote.

distribution is predicted using experimentally derived distribution coefficients to tissue components, such as albumin and phospholipids. The model introduces a minimal mechanistic kidney compartment to predict the active re-uptake of PFOA in the proximal tubules and a mechanistic transporter-mediated entero-hepatic circulation. Furthermore, given the high affinity of PFOA to albumin, the unbound fraction determined in plasma, is adjusted to the differential albumin concentration between the plasma and tissue or tubular fluid compartments, to better describe an albumin facilitated transport of PFOA. The predicted Area Under the Curve (AUC) and half-life were compared to the concentration over time curve from a human volunteer study as well as to observed plasma concentrations and half-lives from the latest human biomonitoring data. Finally, the Global Sensitivity Analysis (GSA) workflow recommended by the OECD PBK guideline was applied to both evaluate the validity of the model and inform about the key parameters determining the PBK model predictions. The newly developed PFOA PBK model can be extrapolated to other PFASs for which these key parameters are already available. The subsequent derivation of these parameters for other PFASs will facilitate the extrapolation of the present PBK models to them as well and enable their human toxicological risk assessment.

Keywords: PFOA, Physiologically Based Kinetic modelling, read-across, Global Sensitivity Analysis, Next Generation Risk Assessment

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) represent a substantial chemical class, comprising thousands of structurally diverse chemicals, which only have in common one fully fluorinated methyl or methylene carbon atom. PFAS are ubiquitous environmental pollutants of anthropogenic nature and therefore of potential risk to human health. Human toxicological risk assessment of PFAS is challenging, due to their sheer number, structural diversity and paucity of the relevant data required to identify and characterize them. A solution to this challenge lies in the application of read-across approach, which involves the extrapolation of information from a data-rich PFAS to make predictions about a similar data-poor one. One example of this approach could be the application of a Physiologically Based Kinetic (PBK) model from PFOA, to inform the development of PBK models for PFASs with analogous physicochemical properties. Despite the availability of several PBK models for PFOA, all of them rely on PFOA-specific animal-sourced data and are optimised to correctly predict the long half-lives observed in humans. This hinders their use in the read-across scope, where PBK models should be as mechanistic as possible, with key toxicokinetic processes are described based on either in vitro or in silico studies. The present study introduces a human mechanistic PBK model for PFOA, that uses only in vitro derived input parameters.

Too many PFAS PFOA toxicokinetic understanding long half-life, persistence strong binding to proteins (especially albumin and fatty-acid binding proteins) and membrane phospholipids Discussion about previous PBK approaches, need of fitting to data & use of parameters taken from animal studies -> read-across not possible - What we need: more generic, that we can use for PFAS for which we don't know information

2. Materials and Methods

2.1. Model structure

The uptake and elimination of PFOA are predicted using in vitro to in vivo extrapolation principles, while its tissue distribution is predicted using experimentally derived distribution coefficients to tissue components, such as albumin and phospholipids. The model introduces a minimal mechanistic kidney compartment to predict the active re-uptake of PFOA in the proximal tubules and a mechanistic transporter-mediated entero-hepatic circulation. Furthermore, given the high affinity of PFOA to albumin, the unbound fraction determined in plasma, is adjusted to the differential albumin concentration between the plasma and tissue or tubular fluid compartments, to better describe an albumin facilitated transport of PFOA.

2.1.1. Mechanistic kidney sub-compartment

2.1.2. Mechanistic liver and intestinal sub-compartments

2.2. Model parametrization

2.2.1. Derivation of partition coefficients

2.2.2. In Vitro to In Vivo extrapolation

2.2.3. Prediction of the adjusted-fraction unbound

2.3. Model evaluation

2.3.1. Global Sensitivity Analysis Workflow

2.3.1.1. Morris screaning test.

2.3.1.2. eFAST test.

2.3.2. Evaluation against human biomonitoring data

3. Results

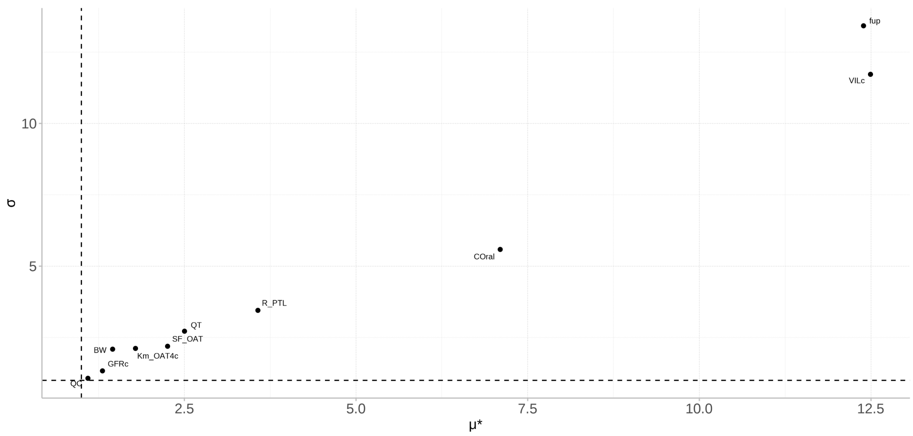
3.1. Model development

Model figure Show results with & without sub compartments, from most complex to less complex and which processess were included

3.2. Model evaluation

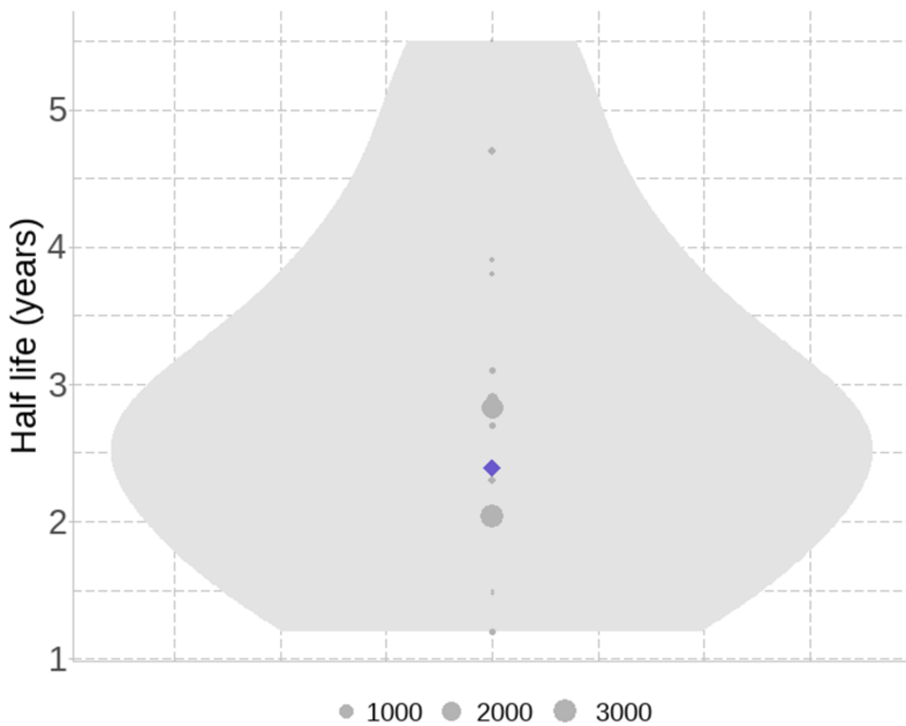
3.2.1. Global Sensitivity Analysis

Finally, the Global Sensitivity Analysis (GSA) workflow recommended by the OECD PBK guideline was applied to both evaluate the validity of the model and inform about the key parameters determining the PBK model predictions.



3.2.2. Evaluation against human biomonitoring data

The predicted Area Under the Curve (AUC) and half-life were compared to the concentration over time curve from a human volunteer study as well as to observed plasma concentrations and half-lives from the latest human biomonitoring data.



4. Discussion

Model limitations : tissue absorption not fully understood, uncertainty of the most sensitive parameters Potential uses?!

The newly developed PFOA PBK model can be extrapolated to other PFASs for which these key parameters are already available. The subsequent derivation of these parameters for other PFASs will facilitate the extrapolation of the present PBK models to them as well and enable their human toxicological risk assessment.

5. Conclusion

6. Funding

7. CRediT authorship contribution statement

8. Declaration of Competing Interest

9. Acknowledgement

10. Appendix A. Supporting information

11. Data availability

12. FYI (for Chrysa)

12.1. Using CSL

If `cite-method` is set to `citeproc` in `elsevier_article()`, then pandoc is used for citations instead of `natbib`. In this case, the `csl` option is used to format the references. By default, this template will provide an appropriate style, but alternative `csl` files are available from https://www.zotero.org/styles?q=elsevier. These can be downloaded and stored locally, or the url can be used as in the example header.

13. Equations

Here is an equation:

$$f_X(x) = \left(\frac{\alpha}{\beta}\right)\left(\frac{x}{\beta}\right)^{\alpha-1} e^{-\left(\frac{x}{\beta}\right)^{\alpha}}; \alpha, \beta, x > 0.$$

Inline equations work as well: $\sum_{i=2}^{\infty}\{\alpha_i^{\beta}\}$

14. Figures and tables

Figure 1 is generated using an R chunk.

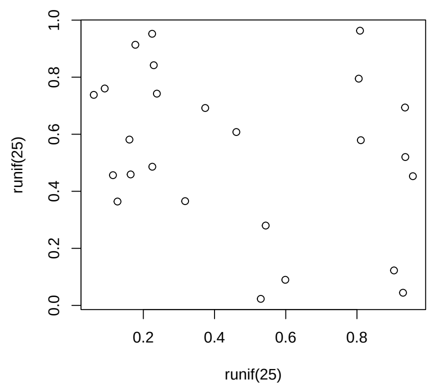


Figure 1: A meaningless scatterplot

15. Tables coming from R

Tables can also be generated using R chunks, as shown in Table 1 example.

```
knitr::kable(head(mtcars)[,1:4])
```

Table 1: Caption centered above table

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

References